Generative or Discriminative?

Revisiting Text Classification in the Era of Transformers

Kasa et al. (Amazon) · EMNLP 2025, Outstanding Paper

CS F429 Paper Presentation · Paper #13 · Group 13

ENC MLM AR AR_pseudo DIFF

Does the classic “two regimes” rule survive transformers?

Generative

Models the text and the label together:
$P(X, y) = P(X \mid y) \cdot P(y)$

- Classic example: Naive Bayes
- Learns fast from few examples
- But a higher error once data is plentiful

Discriminative

Models only the label given the text:
$P(y \mid X)$

- Classic example: logistic regression
- Needs more data to get going
- But a lower error in the end

Known so far: Ng & Jordan (2001) proved this trade-off for linear models; Yogatama et al. (2017) saw it with LSTMs. **Open:** does it hold for modern transformer classifiers, and does it depend on model size?

Five ways to turn a transformer into a classifier

Model	What the model sees	Trained to	How it predicts	Passes
ENC	`[CLS] text`	predict y (cross-entropy)	linear head on [CLS]	1
MLM	`[CLS] text [SEP] The label is y`	fill in 15% masked tokens	fill [MASK] in the label slot, label tokens only	1
AR	`y text`	predict every next token	pick y with the highest log P(text | y)	one per label
AR_pseudo	`text y`	predict every next token	next token after the text, label tokens only	1
DIFF	`text [SEP] The label is y`	undo random masking (discrete diffusion, SEDD)	denoise the label slot	many steps

Discriminative: ENC. Generative: AR, DIFF. “Pseudo-generative”: MLM and AR_pseudo, which learn a likelihood but read the label off as one prediction.

Why might generative models need fewer examples?

1

More training signal per example

ENC gets one target per example (the label). AR and MLM also predict the text tokens, so one example gives dozens of training targets.

2

Learning the text works like built-in pretraining

Modelling P(X) alongside the label acts as a data-dependent regulariser (Erhan et al., 2010). The paper uses this to explain MLM's strength at high data.

3

But capacity is shared

Modelling the text competes with modelling the label. A small model cannot do both well, so the generative advantage should depend on model size.

4

Calibration should favour ENC

ENC's loss targets P(y | X) directly. The others optimise token likelihoods, so their label probabilities need not be calibrated.

The study: 2,835 training runs, all from scratch

5
models

9
datasets (AG News, SST-2, SST-5, IMDb, …)

7
training sizes, 128 up to the full set

3
depths: 1, 6 and 12 layers

3
random seeds

5 × 9 × 7 × 3 × 3 = 2,835 runs

What they measure

- Accuracy: weighted-F1 at every size and depth (Q1)
- Robustness: randomly drop or substitute X% of tokens; report the noise level that costs 5%, 10%, … of F1 (Q2)
- Reliability: calibration (ECE, MCE) and ordinality (MAE, MSE, unimodality) of the predicted probabilities (Q3)

Main finding: which approach wins depends on model size

1 layer

ENC

Only ENC keeps improving with data and wins at high data. No two regimes.

6 layers

DIFF

DIFF is best with little data; ENC overtakes it as data grows. The classic two regimes hold.

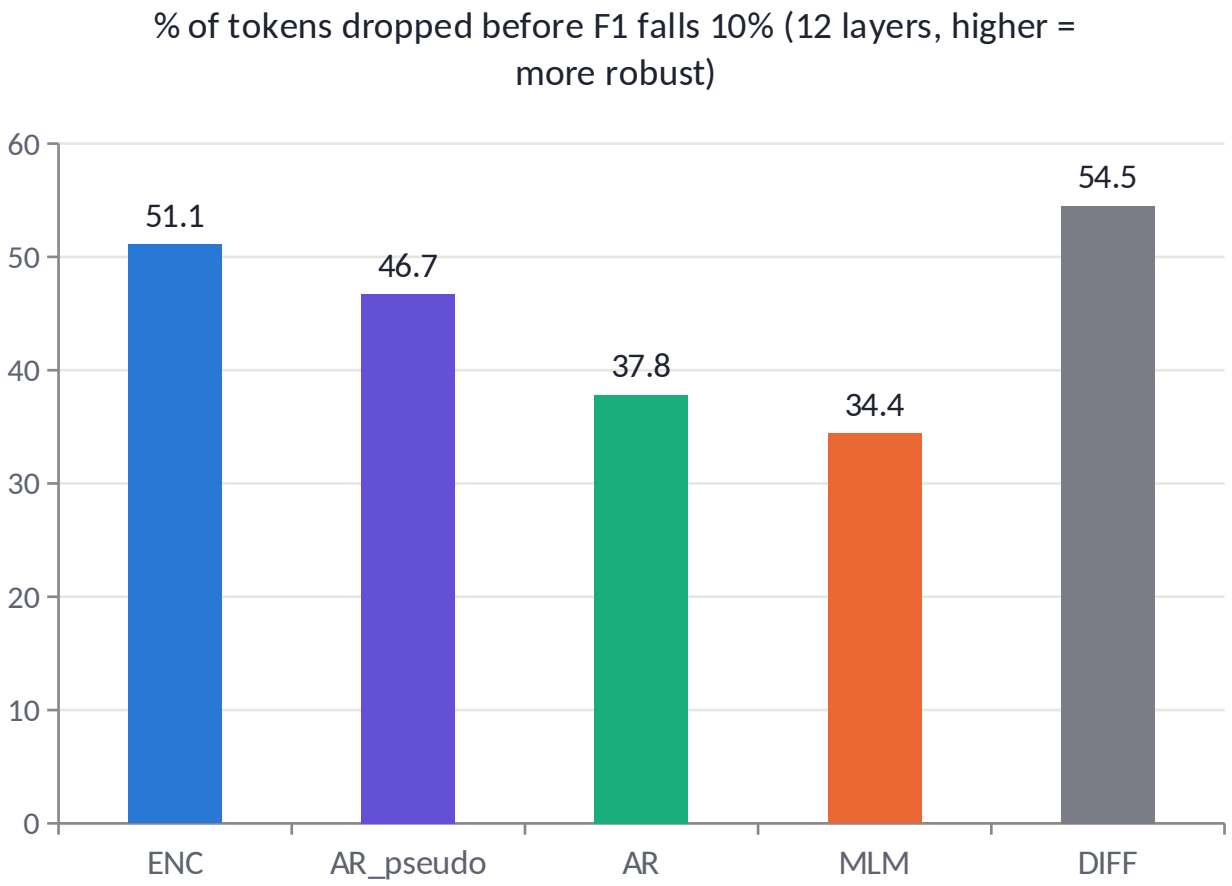
12 layers

AR MLM

AR and DIFF win with little data, the gap shrinks with data, and MLM is best at high data.

Also: AR_pseudo is below AR with little data (and noisier). With pretrained models (BERT vs GPT-2), ENC wins at every size: pretraining removes the two-regimes effect.

Robustness and calibration in the paper



Source: paper, Table 2 (random token dropping).

Robustness

- Substituting tokens hurts every model more than dropping them
- DIFF is the most robust; 12-layer MLM is the least robust to dropping
- AR is sensitive to noise

Calibration

- ENC is well calibrated at every size
- MLM is only as good once data is plentiful
- DIFF gives no usable probabilities, so it has no ECE

The paper's practical advice

ENC **Fast inference needed**

Small 1-layer ENC: efficient, robust to noise, well-calibrated and ordinal outputs.

MLM **Plenty of labelled data, offline**

12-layer MLM: best F1, calibration and ordinality. Careful with noisy input.

DIFF **Little labelled data**

AR or DIFF. DIFF is more robust to noise and strongest at 6 layers, but slow at inference (20 to 100 times ENC).

AR **Calibrated probabilities needed**

Among generative models, AR: better calibrated than AR_pseudo in 7 of 9 datasets at low data.

Limitations the paper does not state

1. Checkpoints chosen on the test set
In the released code, ENC, MLM and AR are validated on the test split (their comment: “for illustration purposes only”). If the paper's runs did the same, scores are optimistic.

2. Unequal training budgets
Appendix B: ENC 30 epochs, MLM 200, AR up to 100. The code sets 200, 200 and 500. Gaps may partly reflect training budget, not the approach.

3. Paper text and code disagree
MLM template (“The label is” vs “Label:”) and AR scoring (log P(X | y) vs mean loss over label and text, which adds a learned class prior).

4. No significance tests
Three seeds and shaded standard deviations only. Many curves overlap, so several “wins” may be within noise.

5. Narrow evaluation data
Short English texts, mostly sentiment. SST-2 is scored on its 872-sentence dev set. Noise is synthetic, at full data only.

6. Gaps in the comparison
DIFF, recommended for low data, has no calibration numbers. The pretrained check covers only ENC vs AR.

Open problems and future work

1. **Compute-matched comparison**
   Give every model the same training budget and tuning, then re-ask the question.

2. **Real shifts, real noise**
   Distribution shift, typos and OCR errors instead of random token edits.

3. **Pretrained MLM and diffusion classifiers**
   Does MLM still win at high data when it starts from pretrained weights?

4. **Fix AR_pseudo's weak label signal**
   The label is one token out of dozens in its loss; up-weighting it is a one-line change.

5. **Hybrid models and theory**
   Generative and discriminative heads on one backbone; an explanation for why depth flips the result.

Our reproduction

Four classifiers, one small transformer, trained from scratch on CPU

What we built, and what we cut

	Paper	Ours
Models	ENC, MLM, AR, AR_pseudo, DIFF	ENC, MLM, AR, AR_pseudo (no DIFF: too slow on CPU)
Size	1, 6, 12 layers; hidden 64 to 768	1 and 4 layers; hidden 128, 4 heads
Tokens	WordPiece / GPT-2 BPE, up to 512 tokens	3,000 most frequent words, first 40 tokens
Data	9 datasets; 128 examples up to the full set	AG News and SST-2; n = 128, 512, 2,048
Training	Grid search per model; 30 to 500 epochs; 3 seeds	One learning rate (0.001); 12 to 40 epochs; 2 seeds
Checkpoint	Test split (released code)	200 held-out training examples; test used once
Noise	Sweep of noise levels, full data, 6 and 12 layers	One level (30%), n = 2,048, 500 test examples

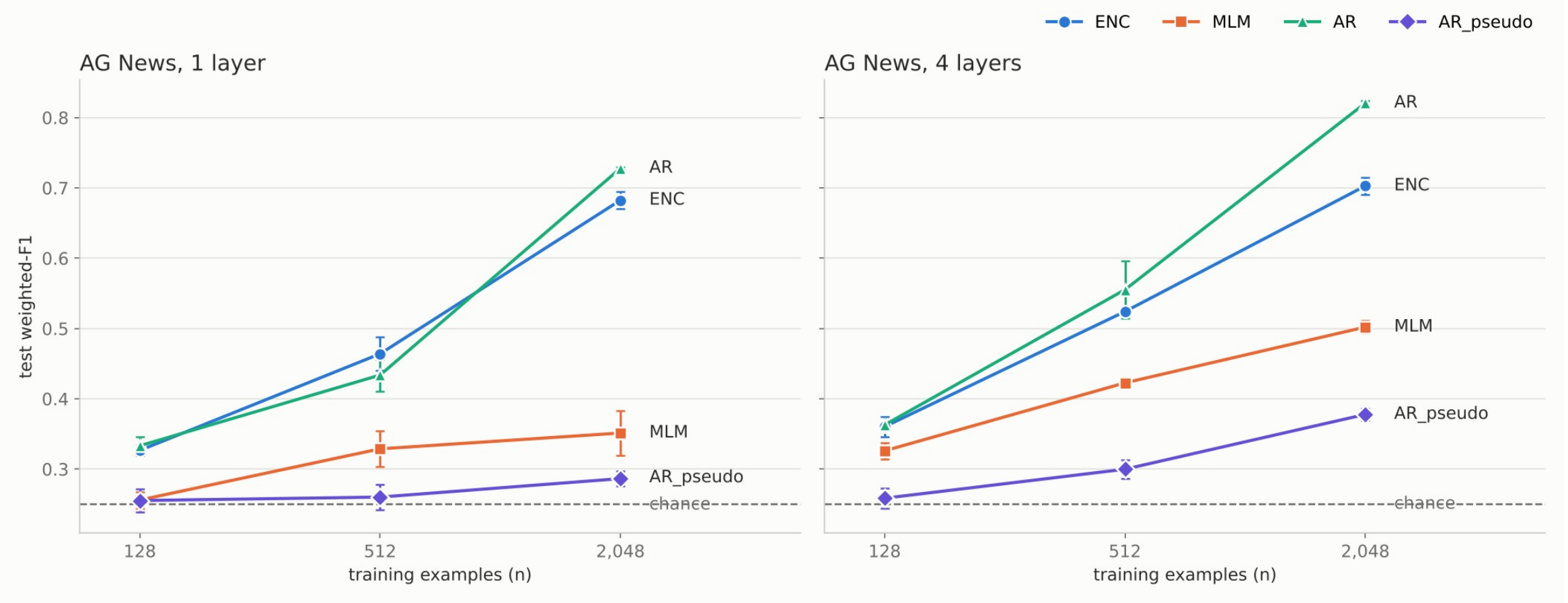
96 runs in about 2 CPU-hours. The mechanisms are faithful; the scale is not.

Where our models differ from the paper and its code

	Same as the paper	Different
AR	Matches the paper's text: score log P(text | y), uniform prior.	Differs from the code, which scores label + text with a mean loss. So our AR probabilities are sharper, and our AR ECE is not comparable to theirs.
MLM	Template “the label is [MASK]” as in the paper.	Differs from both: we mask the label slot in every training example; the paper masks it only when it falls in the random 15%. Our MLM gets a stronger label signal.
AR_pseudo	Label after the text, read out over label tokens.	No “label:” cue word before the label (the code has one).
All models	Same backbone and optimiser for every model.	Labels are dedicated vocabulary tokens, not words.

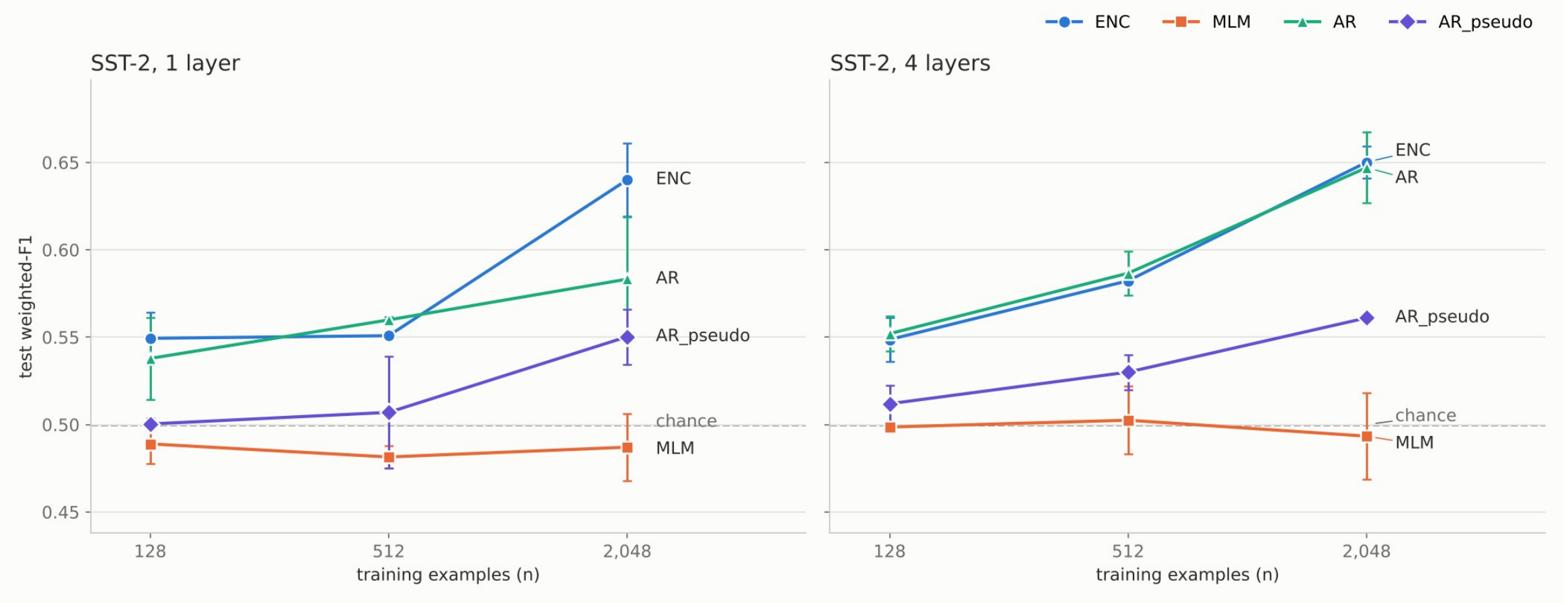
We kept these choices on purpose and report them; changing them would be a separate reproduction.

AG News: AR pulls ahead as the data grows



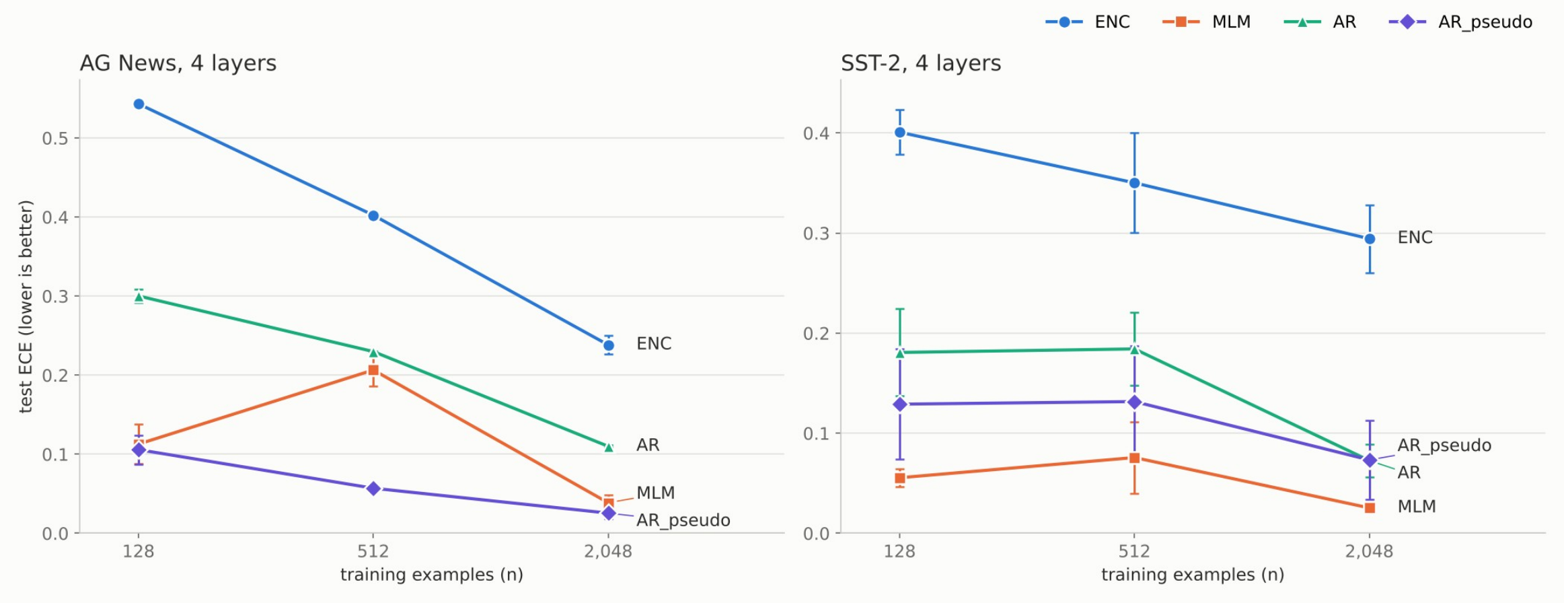
4 layers, n = 2,048: AR 0.821, ENC 0.703, MLM 0.502, AR_pseudo 0.378. At n = 128 AR and ENC are tied (0.363 vs 0.360). Bars show the spread over 2 seeds; chance = 0.25.

SST-2: a weak signal, and ENC and AR tie



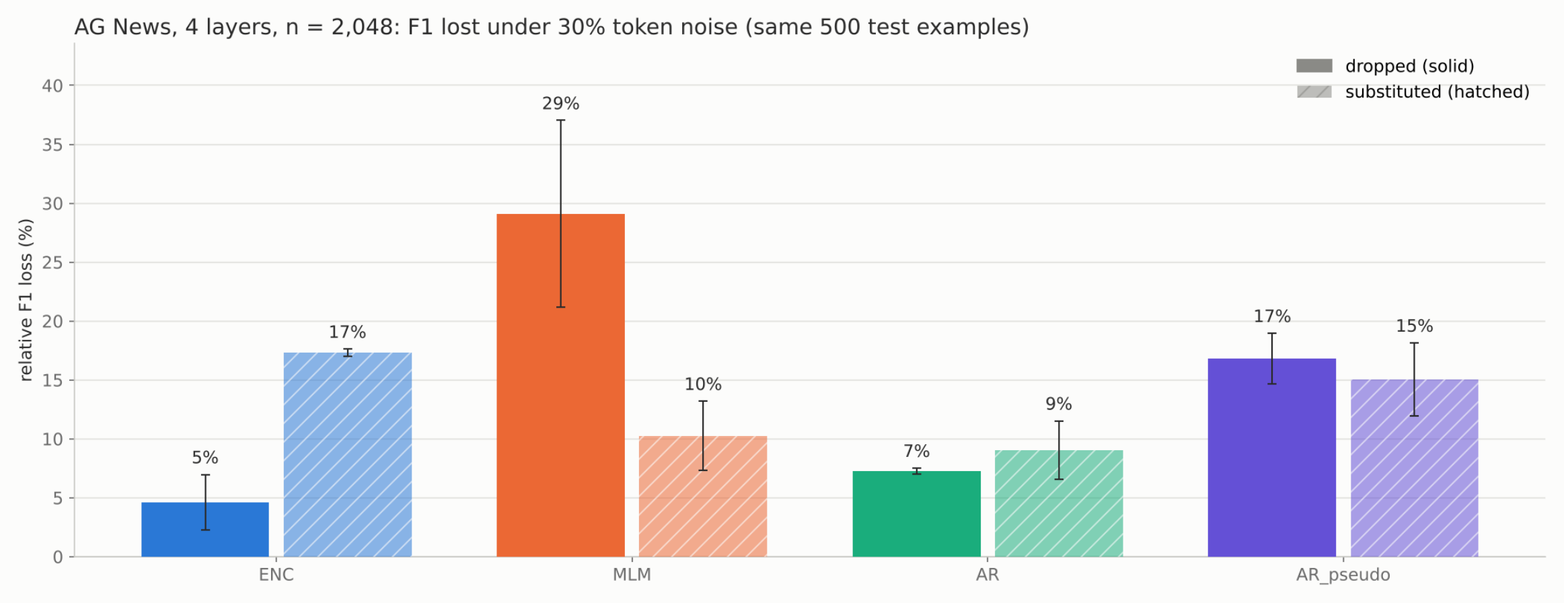
Best result 0.650 (chance 0.5). At 4 layers, n = 2,048: ENC 0.650, AR 0.647; MLM stays at chance. Differences are mostly within seed noise, so SST-2 tells us little at this scale. Scored on the official 1,821-sentence test set.

Calibration: our ENC is the worst, unlike the paper's



AG News, 4 layers, n = 2,048 (ECE, lower is better): ENC 0.238, AR 0.110, MLM 0.039, AR_pseudo 0.026. Caution: a model near chance can get a low ECE just by being unsure, as AR_pseudo does here.

Robustness: dropping tokens hurts our MLM most



F1 lost with 30% of tokens dropped / substituted: ENC 5% / 17%, MLM 29% / 10%, AR 7% / 9%, AR_pseudo 17% / 15%. Clean and noisy F1 on the same 500 test examples.

Paper vs us, claim by claim

Paper's claim	Our result (AG News)	Verdict
Generative wins at low data, gap shrinks with data (12 layers)	Tie at n = 128; gap grows to 0.821 vs 0.703 at 2,048 (4 layers)	**Not reproduced**
1 layer: only ENC keeps improving, best at high data	AR 0.728 vs ENC 0.682 at 2,048; our n stops before “high data”	**Can't test**
AR_pseudo is weaker than AR with little data	AR_pseudo far below AR at every n (best 0.378 vs AR 0.821)	**Reproduced**
MLM is best at high data (12 layers)	MLM 0.502, well below AR and ENC	**Not reproduced**
ENC is well calibrated	ENC has the highest ECE (0.238)	**Not reproduced**
MLM is least robust to dropped tokens	MLM loses 29%, the most of any model	**Reproduced**
Substitution hurts more than dropping, all models	True for ENC and AR, not for MLM and AR_pseudo	**Partly**
AR is sensitive to noise	AR loses 7% / 9% (ENC 5% / 17%); different metric	**Not reproduced**

Live demo: one cell of the grid, trained live

```
python repro.py --dataset agnews --layers 4 --sizes 2048 \
    --seeds 0 --models ENC AR --out /tmp/demo.jsonl
```

Expected (deterministic)	weighted-F1	ECE	30% dropped	30% substituted
ENC	0.715	0.226	0.663	0.587
AR	0.817	0.108	0.768	0.772

Same code and seed as our stored grid, so the numbers should match to three decimals.

What it shows

- AR beats ENC on F1 and is better calibrated, from 2,048 examples
- About 7 minutes on one CPU core: start it at the beginning of the talk
- Fallback: demo/expected_output_n2048.txt (output of this exact command)

Takeaways

- The paper: the generative vs discriminative trade-off depends on model size; there is no single rule.
- Our runs: AR is the strongest model on AG News at our scale, but the paper's size-dependent pattern did not reproduce at 1 and 4 layers.
- Our caveat: small models, short training and one learning rate; several gaps are within seed noise.
- With more compute: 6 and 12 layers, DIFF, subword tokens, per-model tuning, 3 seeds and equal training budgets.